

Candidate Pseudolabel Learning

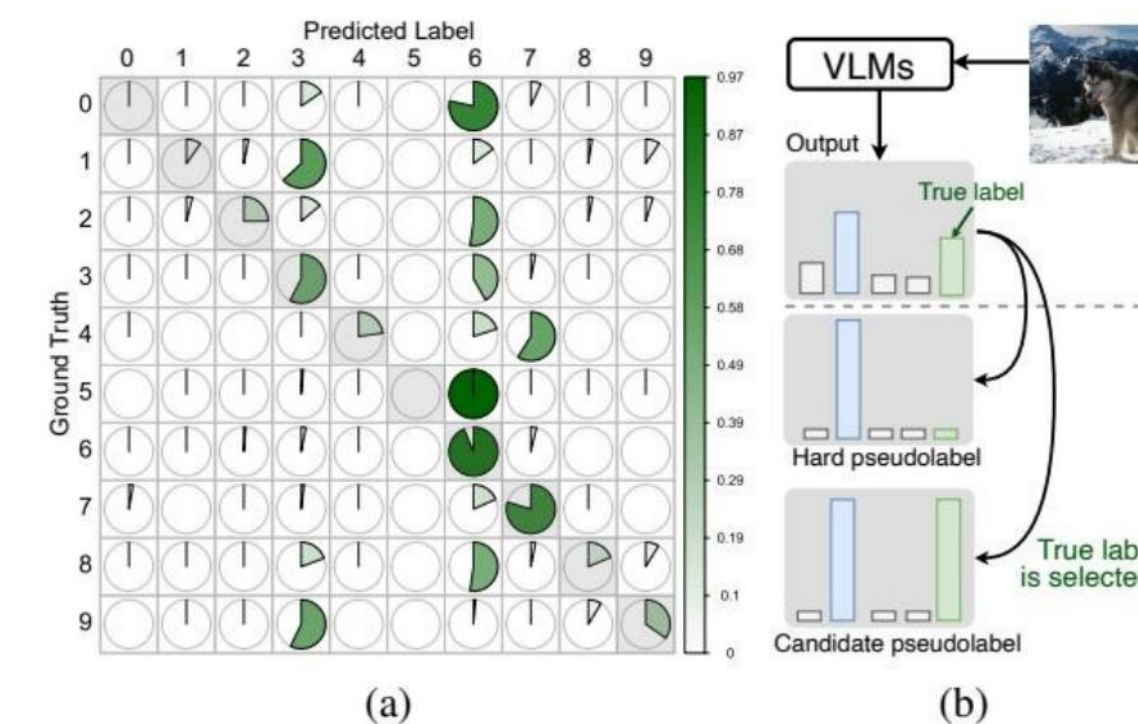
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Why Candidates?

Problem Signal & Intuition

Hard labels amplify noise and class bias in VLMs

Use small candidates, soft targets; robust, PT-ready



Prior Work

Zero-shot CLIP; FPL/GRIP; CPL/LaFTeR; visual prompts

Gaps: single-label; imbalance; unlabeled-scale issues

	Methods	SSL	UL	TRZSL
DTD	Zero-shot CLIP	52.45 _{0.00}		51.61 _{0.00}
	FPL ✗	60.61 _{1.56}	52.99 _{0.43}	60.77 _{0.54}
	CPL ✗	62.78 _{0.17}	57.23 _{0.19}	62.52 _{1.38}
	GRIP ✓	60.91 _{0.00}	54.40 _{0.00}	64.92 _{0.00}
	CPL ✓	69.82 _{0.32}	57.20 _{0.45}	71.97 _{0.46}
RESISC45	Zero-shot CLIP	62.67 _{0.00}		62.13 _{0.00}
	FPL ✗	79.01 _{0.55}	70.85 _{0.66}	77.69 _{0.83}
	CPL ✗	80.38 _{0.37}	76.01 _{0.19}	79.97 _{0.77}
	GRIP ✓	81.53 _{0.00}	76.86 _{0.00}	86.88 _{0.00}
	CPL ✓	87.75 _{0.29}	80.88 _{0.86}	89.73 _{1.73}
Flowers102	Zero-shot CLIP	73.98 _{0.00}		73.05 _{0.00}
	FPL ✗	89.07 _{0.94}	77.81 _{0.30}	91.84 _{0.73}
	CPL ✗	88.37 _{0.39}	82.98 _{0.14}	96.65 _{0.08}
	GRIP ✓	94.21 _{0.00}	82.33 _{0.00}	96.18 _{0.00}
	CPL ✓	96.80 _{0.63}	83.94 _{0.69}	97.34 _{0.74}

Research Requirements and Goals

- Realism: robust to noisy, imbalanced unlabeled data
- Scale: parameter-efficient with minimal supervision
- General across datasets and scenarios
- Simple selection-training recipe for prompt tuning
- Improves text and visual prompts as data grows

Task & Tuning

Paradigms & Backbones

Methods	Flowers102			RESISC45			DTD			
	SSL	UL	TRZSL	SSL	UL	TRZSL	SSL	UL	TRZSL	
Zero-shot CLIP		63.67 _{0.00}	63.40 _{0.00}		54.48 _{0.00}	54.46 _{0.00}		43.24 _{0.00}	43.45 _{0.00}	
FPL	✗	67.03 _{0.65}	65.50 _{0.41}	71.94 _{0.00}	65.14 _{0.25}	62.24 _{0.22}	67.85 _{0.00}	47.60 _{1.09}	47.69 _{0.48}	52.43 _{0.00}
CPL (Ours)	✗	70.58 _{0.13}	68.94 _{0.16}	78.13 _{0.31}	68.85 _{0.13}	67.97 _{0.17}	72.18 _{0.27}	52.64 _{0.68}	50.37 _{0.46}	55.90 _{0.69}
GRIP	✓	67.95 _{1.2}	63.09 _{0.56}	77.18 _{0.00}	71.22 _{0.77}	68.43 _{0.61}	82.19 _{0.00}	54.57 _{4.86}	50.51 _{0.99}	62.78 _{0.00}
CPL (Ours)	✓	73.52 _{0.37}	67.25 _{0.41}	80.14 _{0.73}	78.46 _{0.74}	72.97 _{0.58}	86.67 _{0.33}	58.74 _{0.81}	53.42 _{0.56}	65.31 _{0.78}

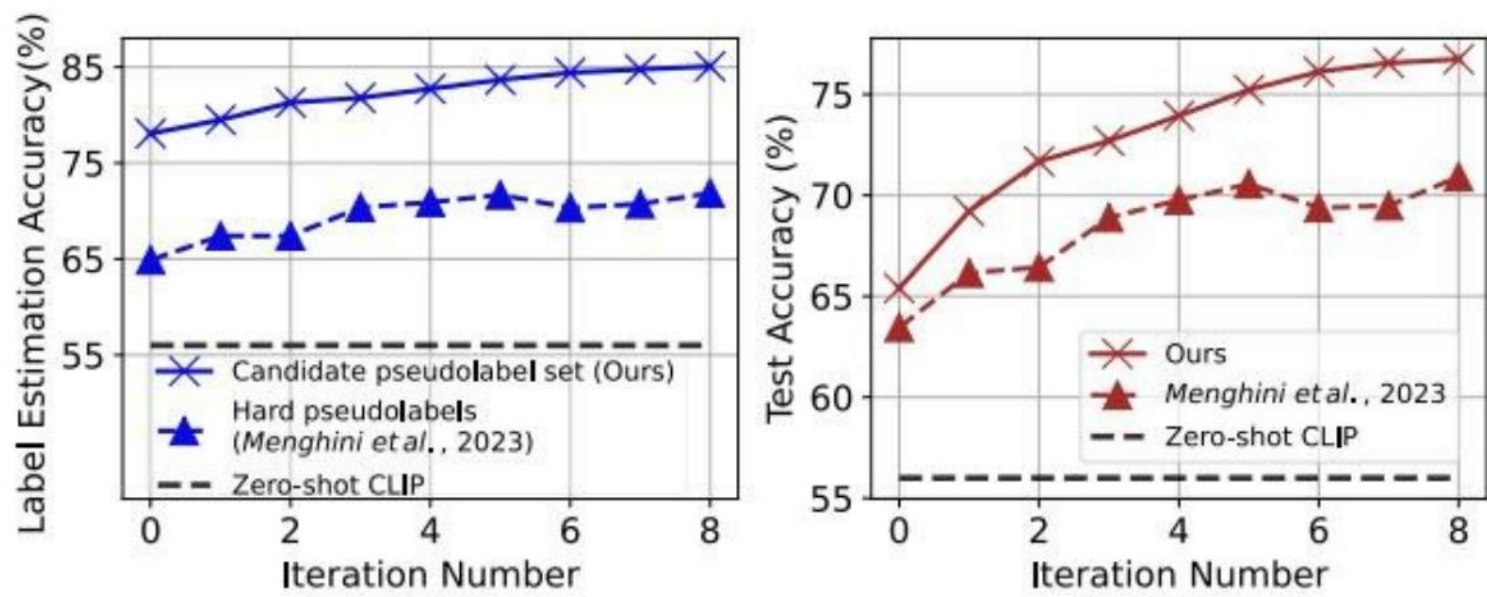
Methods	CUB			EuroSAT			FGVCAircraft			
	SSL	UL	TRZSL	SSL	UL	TRZSL	SSL	UL	TRZSL	
Zero-shot CLIP		51.82 _{0.00}	51.57 _{0.00}		32.88 _{0.00}	30.54 _{0.00}		17.58 _{0.00}	17.86 _{0.00}	
FPL	✗	52.86 _{0.42}	53.17 _{0.06}	54.17 _{0.16}	52.47 _{2.53}	48.79 _{3.69}	68.68 _{14.74}	20.14 _{0.26}	18.28 _{0.33}	16.28 _{0.45}
CPL (Ours)	✗	53.37 _{0.55}	53.28 _{0.31}	56.43 _{0.21}	66.37 _{2.10}	52.83 _{2.10}	74.02 _{1.34}	21.52 _{0.68}	20.10 _{0.51}	26.73 _{0.08}
GRIP	✓	53.83 _{0.11}	52.91 _{0.26}	54.92 _{0.17}	63.48 _{3.09}	63.68 _{3.42}	96.97 _{0.77}	19.43 _{0.50}	17.51 _{0.61}	26.42 _{0.30}
CPL (Ours)	✓	49.50 _{0.42}	52.11 _{0.24}	56.37 _{0.06}	72.03 _{1.24}	68.93 _{1.15}	98.31 _{0.18}	20.51 _{0.68}	18.26 _{0.38}	30.26 _{0.46}

Training Pipeline

From Candidates to Targets

Use unlabeled non-empty candidates; CE + cand. loss.

Iteratively tune prompts; reselect candidates; gains.

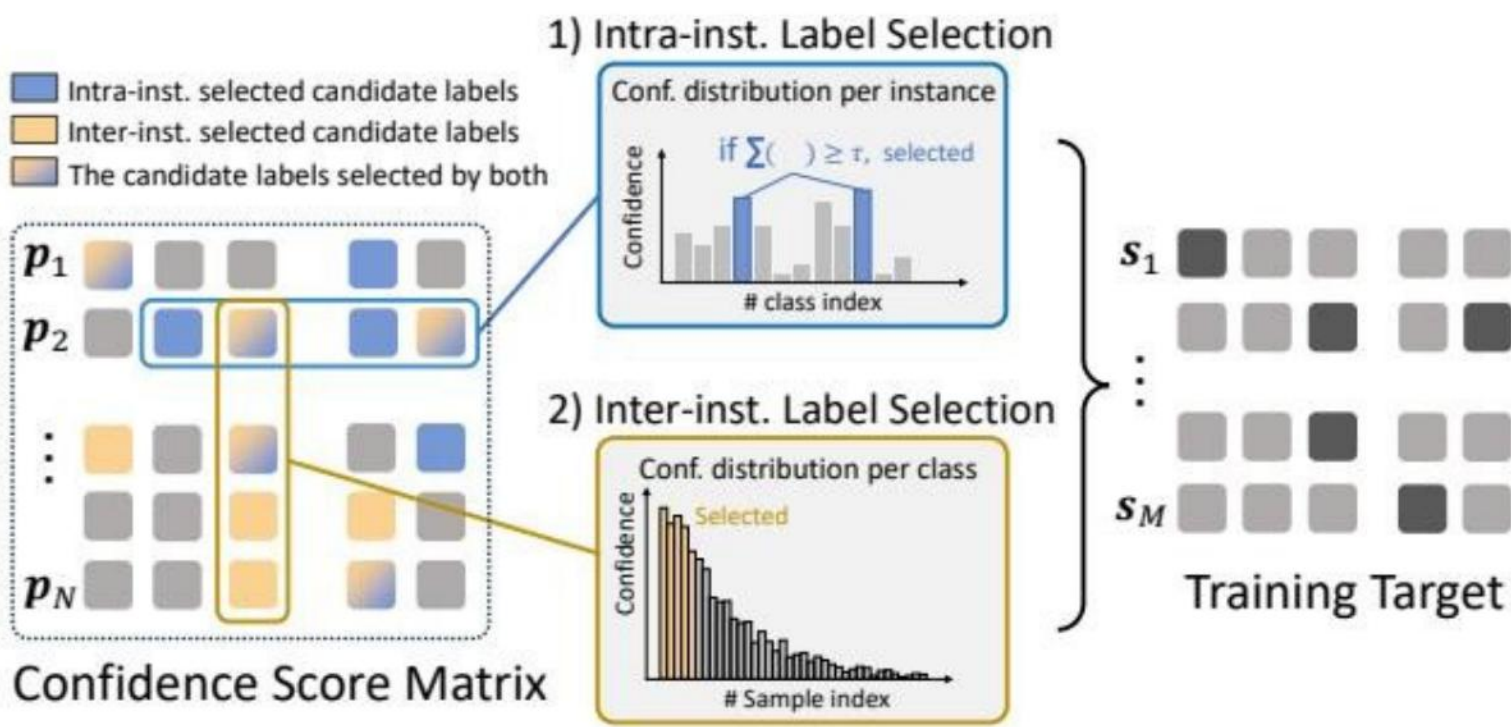


CPL Selection

Two-step Intersection

Intra-instance: keep top labels via adaptive scores.

Inter-instance: quantile filter fights skew; renorm.



Challenges and CPL's Fixes

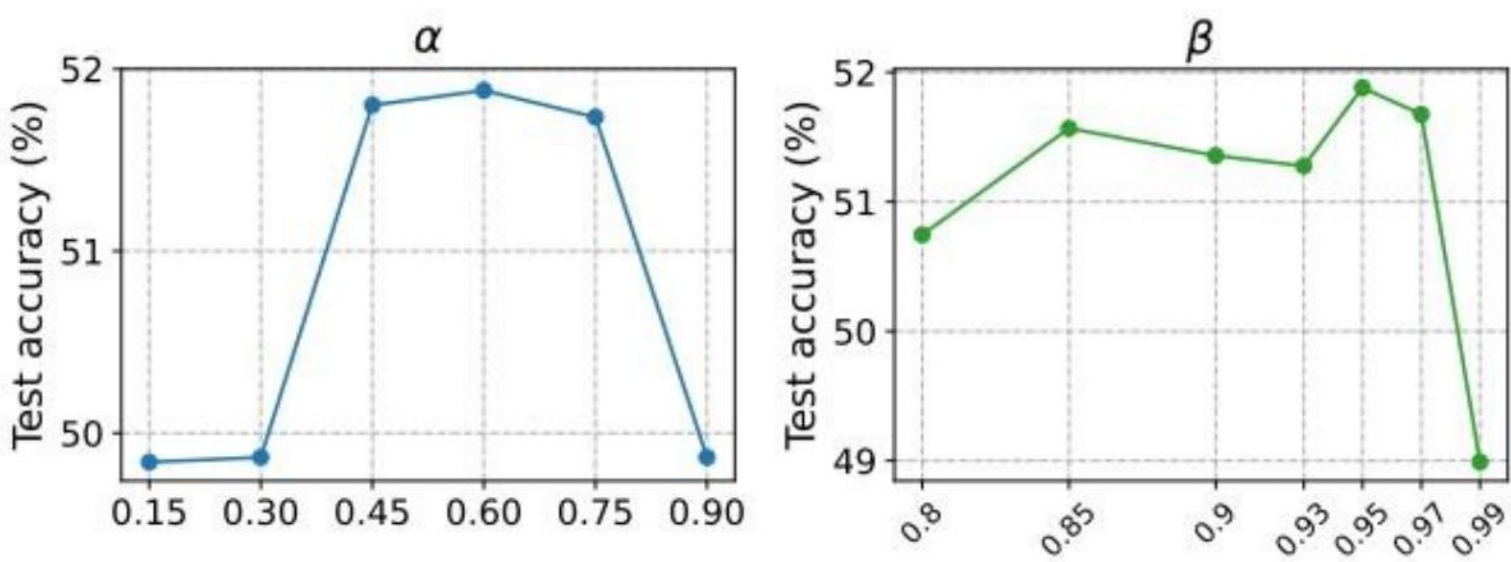
Hard argmax brittle under noise imbalance

Soft candidate targets replace argmax

Alpha: per-instance thresholding

Beta: inter-instance rebalancing

Sensitivity: broad stable regions; easy tuning



Scaling & Data

Datasets, Scenarios, Setup

Six datasets: fine-grained, texture, scene, satellite

Consistent SSL/UL/TRZSL; fixed selection+updates

Table 6. Detailed settings for experiments in Section 7						
	Flowers102	RESISC45	DTD	CUB	EuroSAT	FGVCAircraft
Statistic data						
Class number	102	45	47	200	10	100
Training set size	2040	6300	3760	5594	27000	6667
Testing set size	6149	25200	1880	5794	5000	3333
Training procedure						
Network	ViT-B / 32					
Batch size	64					
Epoch	50 where first two epochs are set for warmup					
Optimizer	SGD					
Momentum	0.9					
Learning rate (LR)	0.02					
Weight decay	5e-2					
LR scheduler	CosineAnnealingLR					
Hyperparameters						
α in intra-instance label selection	0.60	0.90	0.75	0.75	0.75	0.90
β in inter-instance label selection	0.99	0.97	0.95	0.99	0.80	0.97

Ablation & Sens.

What drives CPL gains?

Intra- and inter-instance filters are both vital.

Soft targets beat hard/equal; tuning is robust.

Methods	EuroSAT	DTD	Flowers102
CPL _{CC}	67.26	51.91	72.90
CPL _{RC}	66.38	51.58	71.91
CPL _{CAV}	66.91	49.31	72.06
CPL _{LW}	67.15	52.23	72.33
CPL _{Soft CE}	56.85	49.53	71.87
CPL _{Soft CC}	65.58	50.98	72.58
CPL _{Soft RC}	65.89	50.83	71.89
CPL _{Soft CAV}	64.90	48.78	71.72
CPL _{Soft LW}	65.74	51.04	72.15

Results Overview

Gains and Scaling

Beats CLIP and PSL in unlabeled, semi, transductive.

Param-efficient; +LaFTeR; robust to imbalance; scales.

Methods	Flowers102			RESISC45			DTD		
	SSL	UL	TRZSL	SSL	UL	TRZSL	SSL	UL	TRZSL
Zero-shot CLIP	63.67 _{0.00}			54.48 _{0.00}			43.24 _{0.00}		
FPL	75.96 _{0.74}	65.67 _{0.23}	80.97 _{0.00}	68.13 _{0.55}	63.07 _{0.38}	72.11 _{0.00}	37.10 _{5.45}	44.96 _{0.55}	46.30 _{0.03}
CPL (Ours)	77.36 _{0.24}	70.01 _{0.21}	84.60 _{0.10}	71.73 _{0.57}	68.47 _{0.34}	72.16 _{0.26}	54.63 _{0.79}	48.92 _{0.17}	59.79 _{1.32}
GRIP	83.60 _{0.48}	69.84 _{1.06}	86.26 _{0.00}	74.11 _{0.68}	70.55 _{0.88}	81.07 _{0.00}	56.07 _{0.85}	46.09 _{1.06}	65.30 _{0.01}
CPL (Ours)	89.66 _{0.36}	72.90 _{0.78}	87.35 _{0.76}	80.98 _{0.11}	77.39 _{0.44}	85.85 _{0.49}	61.21 _{0.56}	51.91 _{0.71}	68.00 _{0.34}

Methods	CUB			EuroSAT			FGVCAircraft		
	SSL	UL	TRZSL	SSL	UL	TRZSL	SSL	UL	TRZSL
Zero-shot CLIP	51.82 _{0.00}			32.88 _{0.00}			17.58 _{0.00}		
FPL	55.29 _{0.59}	53.04 _{0.53}	55.44 _{0.20}	62.05 _{1.64}	48.96 _{1.49}	53.70 _{26.87}	20.02 _{0.77}	16.62 _{0.67}	17.55 _{0.37}
CPL (Ours)	56.37 _{0.45}	54.18 _{0.05}	64.01 _{0.17}	64.84 _{2.15}	51.45 _{1.97}	54.03 _{2.27}	22.37 _{0.66}	18.90 _{0.20}	28.47 _{0.43}
GRIP	56.65 _{0.33}	51.42 _{0.21}	59.48 _{0.38}	58.66 _{2.64}	57.21 _{1.77}	92.33 _{0.69}	16.98 _{0.82}	15.22 _{0.71}	26.08 _{0.25}
CPL (Ours)	58.53 _{0.24}	53.47 _{0.36}	66.20 _{0.50}	77.51 _{0.80}	67.26 _{0.47}	93.78 _{0.12}	22.48 _{0.63}	18.35 _{0.27}	30.86 _{0.70}

Final Takeaways: Candidate Pseudolabels

- Balanced candidate sets with soft targets
- Higher accuracy and robustness in prompt tuning
- Dual selection: instance precision + dataset balance
- Parameter-efficient; complements visual prompts
- Scales with unlabeled data; stable hyperparams